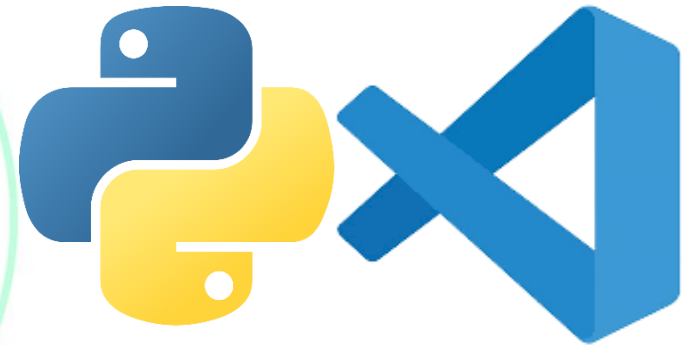


Control Statements in Python (if statement)



With Practical

Outcomes of this Lecture

Sequential Execution

Need of Control Statements

Categories of Control Statements

Importance of Indentation

Implementation of if statement





What is the need of Control Statements?

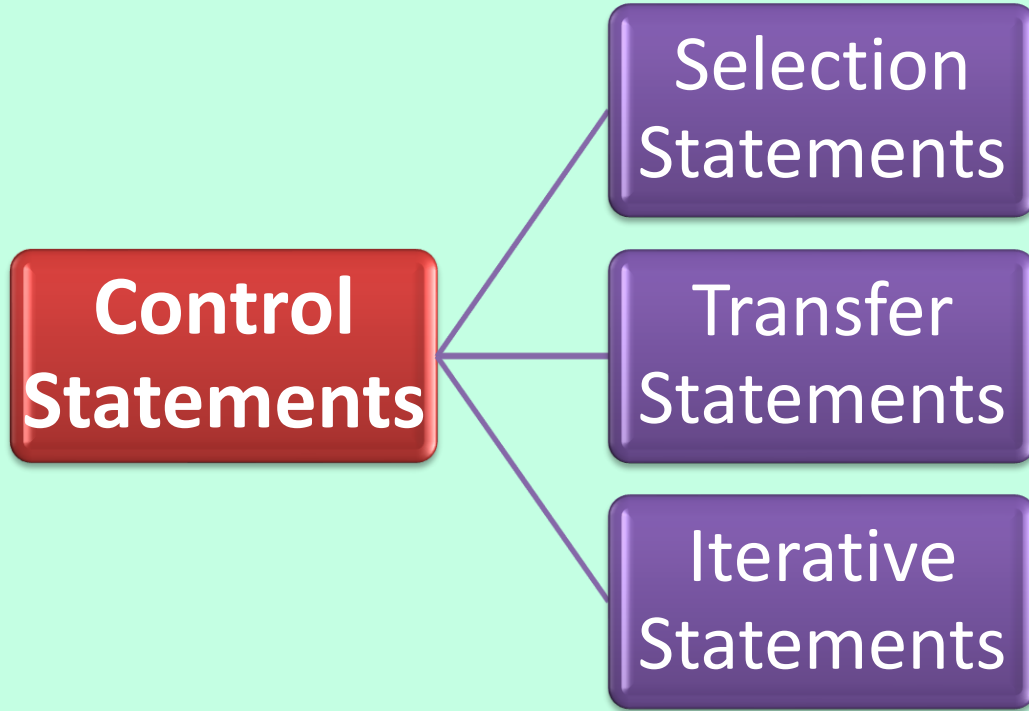
- By default, statements in the program are executed sequentially from the first to the last, is called **“sequential execution”**. It is suitable to develop simple programs.
- To develop critical programs, it may be required to change the flow of execution.
- Flow control describes the order in which statements will be executed at runtime.



Use of Control Statements

- Control statements in Python are used to **control the flow of execution** in a program.
- Instead of executing line-by-line sequentially, they help us make decisions, repeat tasks, or jump between parts of code.

Categories



Note: Selection Statements are also called **Branching Statements** or **Decision Making**

Selection Statements



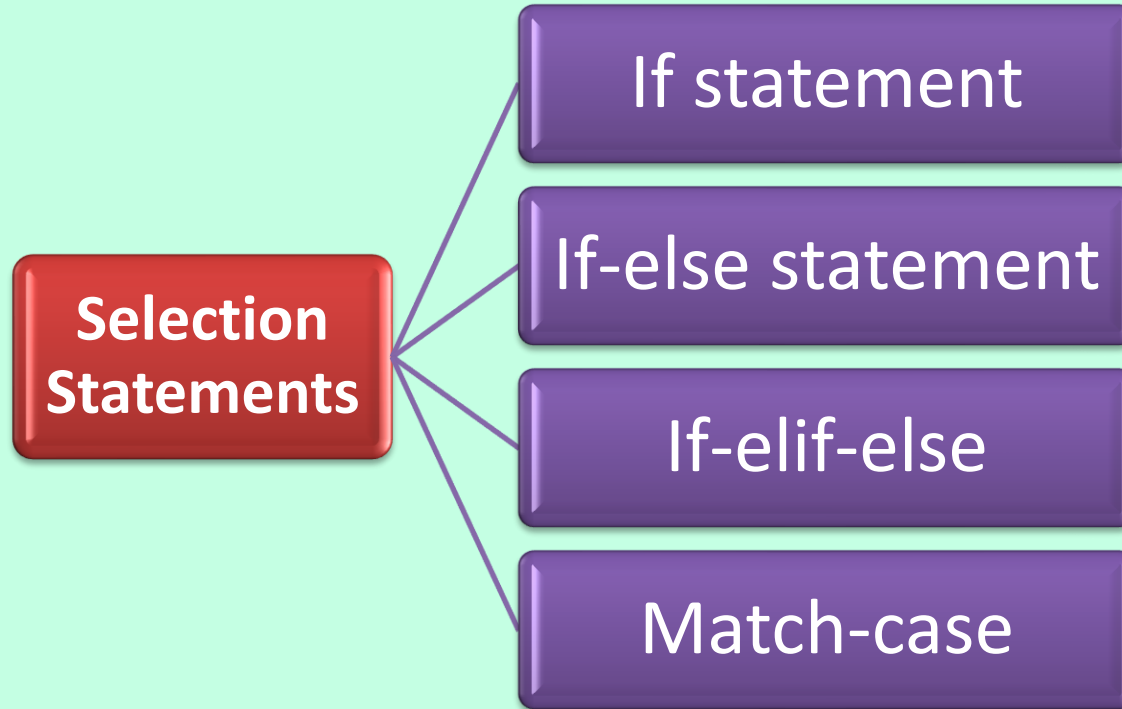
- Used to **make decisions based on condition(s)**
- A block of statement or a block of statements will be executed if a certain expression (condition) is true.
- If condition is false another set of statements will be executed.

Selection Statements Contd..



- The **Comparison/Relational** and **logical operators** are useful in controlling flow of the program.
- Ex: Printing “Good Morning” or “Good Evening” **based on time of the day.**

Types of Control Statements



Note:

The match-case statement was introduced in **2021** with the release of **Python 3.10**



If statement

- If statement is used to execute single statement or block of statements depending on whether the given condition is True or False.
- **Real-life usage:** A website checking if a user has entered a password of at least 8 characters before allowing them to sign up.

Syntax: If statement



- Syntax: **if condition:**

statement-1

statement-2

.....

statement-n

If condition is true
then statements will
be executed.



Indentation

- After the **:** **symbol** start a block with **increased indent**. One or more statements written with the same level of indent will be executed if the given condition evaluates to True.
- To **end the block, decrease the indentation**. Subsequent statements after the block will be executed out of the if condition.

Check your knowledge

What is Sequential Execution?

What is the need of Control Statements?

What are the categories of control statements?

What is the use of Indentation in Python?

How to write python program to find largest of three given numbers?

